Master Informatique / Programmation Système

UFR Sciences Exactes et Appliquées

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The Unisex Bathroom Problem

A *unisex bathroom* is shared by men and women. A man or a woman may be using the room, waiting to use the room, or doing something else. They work, use the bathroom and come back to work. The rule of using the bathroom is very simple: there must never be a man and a woman in the room at the same time; however, people with the same gender can use the room at the same time.

Man Thread
```
void Man(void)
{
   while (1) {
      // working
      // use the bathroom
}
```

Woman Thread
```
void Woman(void)
{
   while (1) {
      // working
      // use the bathroom
}
```

Exercise 1: (50 points) Create a program unisex.c **/** Declare semaphores and other variables with initial values, and add `Wait()` and `Signal()` calls to the threads so that the man threads and woman threads will run properly and meet the requirement. Your implementation should not have any busy waiting, race condition, and deadlock, and should aim for maximum parallelism.
A convincing correctness argument is needed. Otherwise, you will receive no credit for this problem.

Specifically, the program takes two optional command line parameters (the number of females and males, respectively), then creates a thread for each person and simulates bathroom use under some highly contentious conditions...

Exercise 2: (20 points) Create a program unisex1.c **/** The code in *unisex.c* ensures that females and males do not enter the bathroom at the same time but does not restrict the number of people in the bathroom. *unisex1.c* is a copy of *unisex.c* with logic added to take an extra command line parameter specifying the number of bathroom stalls s. Complete the code to enforce the
additional constraint that no more than s women or men can be in the bathroom at any given time.

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Exercise 3: (30 points) Create a program unisex2.c **/** Further modify the code to give priority to women. If there are any men in the restroom and a woman wants to use it, no more men must be allowed to enter, and when the last man leaves the restroom the waiting women must be allowed to enter in arbitrary order. Men must only be allowed to enter the restroom when no women are waiting. As in Exercise 1, no more than *s* people must be allowed in the bathroom at any given time. Submit the code as *unisex2.c*.

Exercise 4: (50 points) Create a program unisex3.c **/** A serious problem with the original code and its modification in Exercise 2 is that some people may never get to use the bathroom if members of the opposite sex keep it occupied indefinitely. A fair solution is obtained by enforcing the following rules:

- If women are in the bathroom and at least one man is waiting outside, no other women are allowed to enter before men get their turn. When the last woman exits, up to *s* waiting men (arbitrarily picked) are allowed to enter the bathroom.
- Similarly, if men are in the bathroom and at least one woman is waiting, no other men are allowed to enter and, when the last man exits, up to s waiting women are allowed in the bathroom.

Implement this solution in *unisex3.c*.